Truth tables

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1 Truth tables of wffs

Consider a wff α with n propositional variables, say $p_1, \ldots, p_n$. The truth table of α consists of $n+1$ columns. The first n entries in a row in the truth table of α consists of some truth values $t_i,\ i = 1, \ldots, n$ assigned to the propositional variables $p_i, i = 1, \ldots, n$ respectively. In other words, every row defines a valuation v such that

$$v(p_i) = t_i,\ i = 1, \ldots, n....(*)$$

The $(n+1)$-th entry of the row gives the truth value of α under that valuation.

So, notice

- for the n propositional variables in α, there would be 2^n distinct rows in its truth table
- there can be many valuations satisfying (*), i.e.

$$v(p_i) = t_i,\ i = 1, \ldots, n....(*)$$

Problem??

No, as we have the following simple but important proposition.

Proposition 1.1. If α is a wff involving propositional variables $p_1, p_2, ..., p_n$ and v, v' are two valuations such that $v(p_i) = v'(p_i)$, i=1,2,...,n, then $v(\alpha) = v'(\alpha)$.

Proof. We use induction on the number m of occurrences of logical connectives in the wff α.

- **Base Case:** $m = 0$, which implies that α is a propositional variable and hence $v(\alpha) = v'(\alpha)$, by assumption.
- **Induction Hypothesis:** The proposition holds for any wff involving, say, k connectives.
- **Induction Step:** We prove that $v(\alpha) = v'(\alpha)$, where α has $k+1$ connectives. Now α can be of the following forms.

  1. $\alpha := \neg\beta$.
     Since α has $k+1$ connectives, β has k connectives, and has the same propositional variables as α, of course. So the conditions of the proposition are satisfied for v and the wff β, and thus, using the induction hypothesis, $v(\beta) = v'(\beta)$.
     This implies that
     $v(\alpha) = v(\neg\beta) = \neg v(\beta) = \neg v'(\beta) = v'(\neg\beta) = v'(\alpha)$.

2. $\alpha := \beta \wedge \gamma$.
   Using induction hypothesis on β, γ, we get $v(\beta) = v'(\beta)$ and $v(\gamma) = v'(\gamma)$. So,
   $v(\alpha) = v(\beta \wedge \gamma) = v(\beta) \wedge v(\gamma) = v'(\beta) \wedge v'(\gamma) = v'(\beta \wedge \gamma)$.
   So $v(\alpha) = v'(\alpha)$.
3. $\alpha := \beta \vee \gamma$.
   Exercise.
4. $\alpha := \beta \rightarrow \gamma$.
   Exercise.

$\square$

The proposition says that if two valuations match on all the propositional variables $p_1, p_2, ..., p_n$ occurring in the wff α, these would match on α itself.

- In other words, propositional variables other than $p_1, p_2, ..., p_n$ do not matter while evaluating the truth values of α under different valuations.
- Effectively then, there are only 2^n assignments of truth values to consider. *These are exactly the ones given by the 2^n rows of the truth table of α.*